INF-2700 Mandatory Assignment Nr. 1

Deadline: Wednesday, 25 September, 2019, 23:59

In this assignment, you are going to perform SQL queries on a given database, through a SQLite shell. You are also going to implement a tiny SQL shell in C.

PART 0. Oranizing Your Files

Make a new directory `inf2700`, on your lab PC, your own PC, or both. This is where you store your INF-2700 materials and all your assignments.

Under `inf2700/`, run:

- `git clone git@inf2700-git.cs.uit.no:2019/public.git public`

In `inf2700/public/` you will find course materials that are public to everybody of this course. For example, you can find this document as `inf2700/public/assignment-spec/assignment1.pdf`.

Assume your UIT account is `xzy123@post.uit.no`. You should have already registered as a user at `inf2700-git.cs.uit.no` and created a private project `xyz123`.

Again, under `inf2700/`, run:

- `git clone git@inf2700-git.cs.uit.no:xyz123/xyz123.git xyz123`.

It is important that `inf2700/xyz123/` is the place hosting all your assignments. Create a directory `sql` for the first assignment.

Create three directories under `inf2700/xyz123/sql/`:

- `docs` for your project documents, including your report,
- `scripts` for your SQL scripts (PART I),
- `src` for your C source code, Makefile, README etc. (PART II).

Copy the database file `public/assignment-repo/inf2700_orders.sqlite3` to `inf2700/xyz123/sql/`.

If you are in the directory `inf2700/xyz123/sql/scripts/`, you can work on the database of this assignment with the following command:

- `sqlite3 ../inf2700_orders.sqlite3`

PART I. SQL

In PART I, you are going to familiarize yourself with basic database operations and SQL using SQLite.

Check SQLite documentation for more information on the use of SQLite.

Task 1. Database schema

Describe the `inf2700_orders` database schema.

Task 2. Run the given SQL queries

Run the queries listed below, and for each query, write in your own words a few lines that describe its purpose and how it works.

a)
```
SELECT C.customerName, C.contactLastName, C.contactFirstName
FROM   Customers C;
```

b)
```
SELECT *
FROM   Orders O
WHERE  O.shippedDate IS NULL;
```

c)
```
SELECT C.customerName AS Customer, SUM(OD.quantityOrdered) AS Total
FROM   Orders O, Customers C, OrderDetails OD
WHERE  O.customerNumber = C.customerNumber
       AND O.orderNumber = OD.orderNumber
GROUP BY O.customerNumber
ORDER BY Total DESC;
```

d)
```
SELECT P.productName, T.totalQuantityOrdered
FROM   Products P NATURAL JOIN
       (SELECT productCode, SUM(quantityOrdered) AS totalQuantityOrdered
        FROM   OrderDetails GROUP BY productCode) AS T
WHERE  T.totalQuantityOrdered >= 1000;
```

Task 3. Write your own SQL queries

Solve the following problems in SQL:

1. Retrieve all customers in Norway.
2. Retrieve all *classic car* products and their *scale*.
3. Create a list of incomplete orders (order status is "In process"). The list must contain orderNumber, requiredDate, productName, quantityOrdered, and quantityInStock.
4. Create a list of customers where the difference between the total price of all ordered products and the total amount of all payments exceeds the credit limit. The list must contain the customer name, credit limit, total price, total payment and the difference between the two sums.
5. Create a list of customers who have ordered all products that customer 219 has ordered.

You are encouraged to run as many SQL queries as you like, such as the ones similar to the examples in the text book, though you do not have to report this effort in your hand-ins.

PART II. Database Programming

In PART II, you are going to implement a database application in C.

Task: Implement a database application

Implement an application called `sql-json` in C that runs your own database "shell". In your `sql-json` shell, you can run SQL queries interactively. The query results are presented in JSON format. A "`.quit`" command ends the session. Below is an example session in the shell:

```
$ ./sql-json ../inf2700_orders.sqlite3
Welcome to sql-json
Enter ".help" for instructions
Enter SQL statements terminated with a ";"
Enter ".quit" to exit

sql-json> select orderNumber, orderDate from Orders where status = 'Cancelled';
```

```
[{"orderNumber": 10167, "orderDate": "2003-10-23"},
 {"orderNumber": 10179, "orderDate": "2003-11-11"},
 {"orderNumber": 10248, "orderDate": "2004-05-07"},
 {"orderNumber": 10253, "orderDate": "2004-06-01"},
 {"orderNumber": 10260, "orderDate": "2004-06-16"},
 {"orderNumber": 10262, "orderDate": "2004-06-24"}]
sql-json> .quit
Leaving sql-json
$
```

Hand-in

When your are working on the assignment, you should make frequent commits to your `xyz123` project. Your commits should include the SQL and C source code, Makefile, and README that describes how to run your code. In addition, you must hand in a report that includes your answers and any special observations you made, problems you experienced etc. The report should be in PDF form.

Make sure you have committed and pushed all your final working before the deadline.

Enjoy coding and good luck!